

Airlock

The airlock provides access for EVAs. It can be located in one of several places: inside the Orbiter crew module in the middeck area mounted to the aft bulkhead, outside the cabin also mounted to the bulkhead or on top of a tunnel adapter that can connect the pressurized Spacehab module with the Orbiter cabin. A docking module can also serve as an EVA airlock.

The airlock contains two spacesuits, expendables for two six-hour payload EVAs and one contingency or emergency EVA, and mobility aids such as handrails to enable the crew to perform a variety of tasks. The airlock allows two crewmen room for changing spacesuits.

Midfuselage

In addition to forming the payload bay of the Orbiter, the midfuselage supports the payload bay doors, hinges and tiedown fittings, the forward wing glove and various Orbiter system components.

Each payload bay door supports four radiator panels. When the doors are opened, the tilting radiators are unlatched and moved to the proper position. This allows heat radiation from both sides of the panels, whereas the four aft radiator panels radiate from the upper side only.

The Remote Manipulator System (RMS) is a 15.2-metre-long articulating arm remotely controlled from the flight deck of the Orbiter. The elbow and wrist movements permit payloads to be grappled for deployment out of the payload bay or retrieved and secured for return to Earth.

A television camera and lights near the outer end of the arm permit the operator to see on television monitors what his hands are doing. In addition, three floodlights are located along each side of the payload bay.

Aft Fuselage

The aft fuselage consists of the left and right orbital manoeuvring systems, Space Shuttle main engines, body flap, vertical tail and Orbiter/external tank rear attachments.

The forward bulkhead closes off the aft fuselage from the midfuselage. The upper portion of the bulkhead attaches to the vertical tail. The internal thrust structure supports the three Space Shuttle main engines, low pressure turbo-pumps and propellant lines.

The shuttle has the most reliable launch record of any rocket now in operation. Since 1981, it has boosted more than 1.36 million kilograms (3 million pounds) of cargo into orbit. More than 600 crew members have flown on its mis-